

Fostering responsible AI literacy: A systematic review of K-12 AI ethics education

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ABSTRACT

AI ethics education remains significantly underprioritized in classroom practice, despite the global push for AI literacy in K-12 curricula. This systematic review analyzes 68 peer-reviewed publications (from January 2014 to March 2025) to map the research landscape of K-12 AI ethics education. The results reveal global trends in the current research landscape, pedagogical designs addressing major responsible AI (RAI) principles, and various learning assessment methods, and students' ethical learning outcomes manifested across cognitive, affective, and behavioral domains. By further synthesizing findings, we identify disparities of eastern and western contexts, gaps in addressing RAI principles and emerging ethical issues, limitations of methodological methods, and issues in assessing ethical learning outcomes. Building on this synthesis, we propose a competency-based responsible AI literacy framework that reconceptualizes AI ethics as a transformative learning dimension that progressively permeates all the learning dimensions of AI literacy development. The framework also provides actionable insights to empower K-12 educators and policymakers in fostering students' responsible AI literacy. The review concludes with three future research directions to advance this critical field.

1. Introduction

Artificial intelligence (AI) is transforming nearly every aspect of human society, delivering significant economic and societal benefits (Tomašev et al., 2020), while simultaneously raising complex ethical, legal, and governance challenges (Coeckelbergh, 2020; Qian et al., 2024). AI was first defined in 1956 as the science and engineering of making intelligent machines (Ng et al., 2021). Over the decades, AI has evolved into sophisticated machines and algorithms capable of reasoning and adapting based on rules and inputs to mimic human intelligence. Mikalef and Gupta (2021), which defines AI as a system that can identify, interpret, make inferences, and learn from data to achieve predetermined organizational and societal goals. This highlight AI's socio-technical and socio-cultural nature, emphasizing the interplay between technology, cultural context, and the social systems it influences (Dolata et al., 2022; Kim & Lee, 2024; Sartori & Theodorou, 2022). The rapid advancement of AI has intensified global efforts to incorporate AI education into K-12 curricula in recent years, equipping teachers and students with essential AI literacy and competency (Miao & Shiohira, 2022).

While AI provides efficiency and convenience in our daily lives,

workplaces, and learning environments, its malicious use can result in negative consequences. To address this, researchers have particularly emphasized the importance of AI ethics (Borenstein et al., 2021) and considered it as an essential component of AI literacy (Ng et al., 2021). In the early years, AI ethics seeks to mitigate the negative consequences brought by AI development, such as algorithmic bias, privacy violations, lack of opacity, and risks of manipulation (Coeckelbergh, 2020; Müller, 2020). The recent rise of large language models (LLMs) like ChatGPT, a class of generative AI (GenAI) systems trained on large text corpora to produce human-like language outputs, has further expanded AI's ethical complexities, introducing concerns such as AI hallucinations, copyright infringement, and the erosion of human creativity (Hagendorff, 2024; Peng & Zhao, 2024). Together, these issues underscore the essence to implement AI ethics education, empowering people to critically examine AI's societal implications as responsible digital citizens (Borenstein & Howard, 2021). In K-12 education, the global push of cultivating AI literacy for K-12 students are expected to prepare them with fundamental understanding and awareness about AI ethics (Dabbagh et al., 2024).

However, despite wide recognition on its importance of AI literacy conceptual frameworks (e.g., Chan, 2023; Kong & Zhang, 2021; Ng

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et al., 2021), AI ethics education remains markedly absent from most AI teaching and learning classroom practices worldwide, revealing a critical gap between its theoretical recognition and practical implementation. UNESCO's 2022 analysis of government-endorsed K-12 AI curricula revealed that despite strong policy commitments, AI ethics and societal impacts receive minimal instructional time, averaging 24 % of overall AI teaching content (Miao & Shiohira, 2022). Implementation challenges are evident across diverse contexts: in the United States, some teachers report confusions in delivering ethics-focused curricula, for example MIT's AI ethics middle school program (Miao & Shiohira, 2022; Williams et al., 2021). The disparity is even more pronounced in China, where video analyses of 98 AI literacy lessons show just 5.1 % address AI ethics, while higher-order competencies (Evaluating & Creating AI at 35.71 %) appear much less frequently than basic competencies (Knowing & Understanding AI and Using & Applying AI, both at 93.88 %) (Wu et al., 2024). This pattern culminates in concerning outcomes: an AI education survey conducted in Qingdao, China, found that near-total absence of AI ethics teaching in K-9 classrooms, resulting in 'almost zero' students' awareness of the ethical issue around AI (Gong et al., 2020).

These gaps highlight the urgent need for a systematic review of AI ethics education to inform educators to teach AI ethics in K-12 settings and foster well-round AI literacy. While existing reviews have explored broader aspects of AI education (e.g., Casal-Otero et al., 2023; Li et al., 2024; Liu & Zhong, 2024; Ng et al., 2023), none have comprehensively examined (1) the pedagogical designs for specific AI ethics topics across diverse cultural and curricular contexts, (2) the evaluation methods employed for assessing ethical learning, and (3) the ethical learning outcomes engaged with overall AI literacy development. To address these gaps, this review synthesizes current empirical studies of K-12 AI ethics education.

2. Literature review

2.1. AI literacy

Although there is no consensus in defining AI literacy yet, AI ethics is widely acknowledged as a critical component of AI literacy frameworks (Almatrafi et al., 2024; Yim, 2024a). In the early years, AI literacy or education was viewed from technological and cognitive dimensions. The Five Big Ideas framework pioneered by AI4K-12 experts positions the "Societal Impact of AI", highlighting both positive and negative consequences of AI applications as a central learning dimension, surrounded by four technical dimensions: *Perception, Representation & Reasoning, Learning, and Natural Interaction* (Touretzky et al., 2019). Building on this, Long and Magerko (2020) proposed a working definition of AI literacy as "a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace" (p. 2), implicitly integrating ethical considerations. They further identified ethics as one of the seventeen key competencies under *How Should AI Be Used?*—specifically, Competence 16 (Ethics) – "identify and describe different perspectives on the key ethical issues surrounding AI (i.e. privacy, employment, misinformation, the singularity, ethical decision making, diversity, bias, transparency, accountability)." However, this overlaps with other competencies of other categories, such as Competency 6 (Imagine future AI), which involves envisioning AI's future societal impacts, and Competency 13 (Critically Interpreting Data), which emphasizes questioning data validity.

Subsequent frameworks have further refined the role of AI ethics by explicitly distinguishing it as a distinct domain separate from technical knowledge or skills. Wong et al. (2020) introduced a succinct three-dimensional AI literacy model (i.e., *AI knowledge, AI applications, and AI safety & ethics*), while Chiu (2021) outlined a curriculum design framework encompassing *Knowledge in AI, Process in AI, Impact of AI*. Kong and Zhang (2021) identified three core components—AI concepts,

evaluation, and problem-solving—while emphasizing the need to "protect benefits and privacy" (p. 1), reinforcing ethical considerations. Similarly, Yue et al. (2025) expanded this into a four-dimensional model (i.e., *AI knowledge, skills, ethics, and attitudes*). Other frameworks adopt a competence-based approach but similarly isolate ethics as a critical domain: Ng et al. (2021)' aligned AI literacy with Bloom's Taxonomy (Bloom et al., 1956), structuring competencies into four tiers: *Knowing & understanding AI, Using & applying AI, Evaluating & creating AI, and AI ethics*, while Carolus et al. (2023) conceptualized AI literacy with four similar domains: *Using & applying AI, Understanding AI, Detecting AI, and Ethics*. Ng, Wu, et al. (2024) further extended the digital AI literacy to other dimensions such as affective, behavioral, and ethical learning domains (see Fig. 1) and updated their formerly proposed AI literacy framework proposed in 2021.

However, these conceptualizations share a less examined facet: the ethical dimension, which holds systemic importance yet remains ambiguously intertwined with other domains, whether foundational knowledge (e.g., understanding algorithmic bias), user attitudes (e.g., critically evaluating AI systems), or practical applications (e.g., ethically deploying tools). Recent studies highlight ethics' growing prominence. Chan (2023) defined an AI-literate learner as one who *comprehends AI terminology, uses applications critically, discerns realistic AI capabilities, and prioritizes safety, security, and responsible use*. O'Dea et al. (2024) argued that ethical dimensions should span cognitive domains, highlighting the need for users to understand ethical principles, apply AI applications responsibly, and create ethical artifacts and systems. This needs further effort to specify the exact AI ethical knowledge, skills, and attitudes required for each cognitive phase. To address this, our study adopts Ng, Wu, et al. (2024)'s framework while advancing a key reconceptualization: rather than treating ethics as a compartmentalized domain, we position it as a transformative dimension that actively intersects with and enriches all facets of AI literacy development. This perspective positions ethical considerations as: (1) integral to cognitive understandings (e.g., interrogating training data biases), (2) foundational to affective attributes (e.g., fostering ethical awareness), and (3) inseparable from behavioral engagements (e.g., embedding human values when designing models). It addresses the current limitation by making ethics both more explicit in each learning dimension and more cohesive across the entire AI literacy framework.

2.2. Review studies of K-12 AI education

Although numerous literature reviews on K-12 AI education have been published in recent years, none of them systematically examine AI ethics education as a central learning domain. Existing studies predominantly suffer from two critical limitations: First, while pre-2022 empirical studies on K-12 AI education existed, few specifically investigated ethics components, resulting in superficial treatment that positions AI ethics as a peripheral subtopic rather than documenting what ethical concepts are taught or how they're delivered, as clearly evidenced by their quantitative analyses. For example, Rizvi et al. (2023)'s systematic review of AI teaching and learning identified only 7 empirical studies (2019–2022) addressing socio-ethical aspects, compared to 24 studies on AI applications and 27 on AI models. Similarly, Casal-Otero et al. (2023) found merely 6 articles among their 179 reviewed documents, addressing ethical and societal implications of student learning experiences focused on understanding AI. These marginalizations persists in results sections of these review studies (e.g., Ng et al., 2023; Su et al., 2023; Tan & Tang, 2024; Yue et al., 2022), typically limited to single-paragraph mentions in most of these studies without breaking down into specific ethical topics and teaching pedagogies, despite frequent acknowledgments of ethics' importance in discussion and implications sections, indicating a troubling disconnect between recognized priorities and research findings.

Second, although existing literature has provided comprehensive syntheses of K-12 AI education as a whole, few studies have specifically:

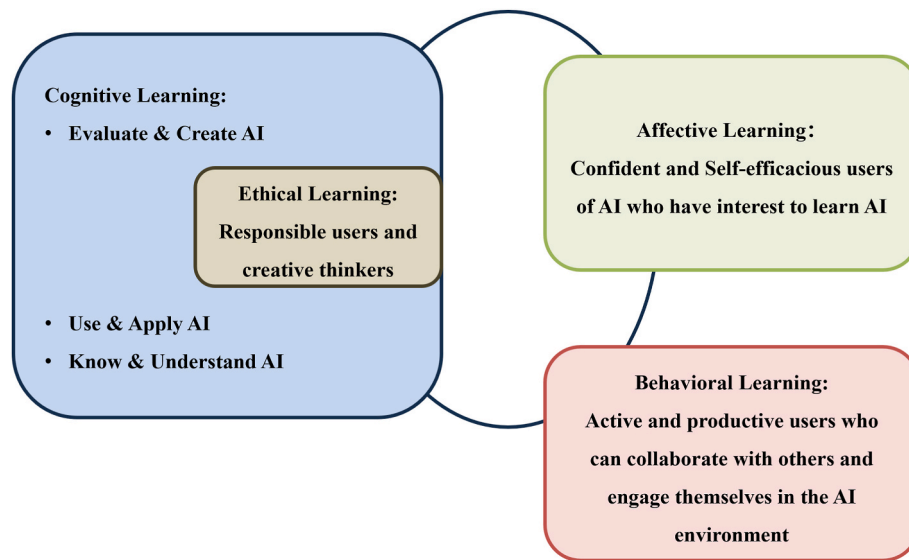


Fig. 1. The ABCE AI literacy Framework modified from Ng, Wu, et al. (2024).

(1) critically evaluated pedagogical designs for teaching AI ethics-related topics, or (2) systematically analyzed how students engage with ethical learning across developmental stages through cognitive, affective, and behavioral dimensions. For instance, Lee and Kwon (2024) identified several ethical issues taught in K-12 classrooms but presented them through overlapping, poorly differentiated categories (i. e., AI ethics and bias, societal considerations, social impacts and challenges of AI, and ethical implications). Although their study explored linkages between ethical learning and critical thinking, its narrow sample of AI ethics studies ($n = 6$) precluded meaningful analysis of how these ethical competencies develop across age groups. On the other hand, Liu and Zhong (2024) advanced this work by systematically mapping AI learning content—including *Knowledge in AI*, *Impact of AI*, *Process in AI*—across grade bands. However, while they analyzed age-appropriate progression of students' learning outcomes, their study did not identify the specific ethical topics taught when briefly summarizing the *Impact of AI* domain. These oversights neglect the nuanced nature of ethical issues surrounding AI, blurring this domain into an undifferentiated whole rather than examining how specific ethical principles (e.g., fairness, transparency) are taught and learned across developmental stages.

2.3. Responsible AI principles

In response to the increasing ethical challenges and societal impacts brought by AI, terms such as 'Ethical AI', 'Trustworthy AI', and 'Responsible AI (RAI)', often used interchangeably (Díaz-Rodríguez et al., 2023; Dignum, 2019; McGovern et al., 2022), have emerged as guiding sets of standards for the research, creation, deployment, and use of AI in ways that aligns with human values. The International Organization for Standardization (ISO) defines responsible AI as "An approach to developing and deploying artificial intelligence from both an ethical and legal standpoint. The goal is to employ AI in a safe, trustworthy and ethical way (ISO, 2023)." RAI extends beyond the discussion of AI ethics by emphasizing the integration of ethical considerations into practical design and implementation, ensuring that AI technologies are developed and applied responsibly (Mikalef et al., 2022; Peters et al., 2020).

Companies and organizations worldwide have established their own RAI guidelines. For instance, OECD (2021) supports RAI through four main pillars: *Inclusive Growth*, *Sustainable Development*, & *Well-being*, *Human-centered Values & Fairness*, *Transparency & Explainability*, and *Robustness, Security, & Safety*. Microsoft released their RAI standard with

six adherences: *Accountability*, *Transparency*, *Fairness*, *Inclusiveness*, *Reliability & Safety*, *Privacy & Security*. UNESCO (2022) outlines ten ethical AI principles to align with the UN's sustainable development goals (SDGs), including *Proportionality & Do No Harm*, *Safety & Security*, *Fairness & Non-discrimination*, *Sustainability*, *Right to Privacy & Data Protection*, *Human Oversight & Determination*, *Transparency & Explainability*, *Responsibility & Accountability*, *Awareness & Literacy*, and *Multi-stakeholder*, *Adaptive Governance & Collaboration*. While these frameworks vary in structure, they converge on core ethical imperatives: ensuring AI benefits society while mitigating risks.

Recently, Papagiannidis et al. (2025) distilled seven core pillars of RAI principles from 48 foundational documents on AI governance, including: *Accountability*, *Human agency and oversight*, *technical robustness and safety*, *Privacy and data governance*, *Transparency*, *Diversity*, *Non-discrimination*, and *fairness*, and *Societal and environmental well-being*. *Accountability* refers to establishing clear responsibility for AI system outcomes and addressing harms. *Human agency and oversight* emphasize maintaining meaningful human control over AI systems. *Technical robustness and safety* require AI to perform reliably under varying conditions with safeguards against failures. *Privacy and data governance* involves ethical collection, use, and protection of personal data throughout the AI lifecycle. *Transparency* means AI systems should be understandable and their decision-making processes explainable. *Diversity*, *non-discrimination* and *fairness* entail preventing biased or exclusionary outcomes in AI systems. *Societal and environmental well-being* consider the broader impacts of AI on communities and ecosystems, including sustainability concerns.

In the education arena, the call-to-actions of introducing RAI in education are gaining momentum (Pargman et al., 2024). Nevertheless, most studies focus on proposing the ethical principles for applying AI in Education (AIED) (e.g., Adams et al., 2023; An et al., 2024; Nguyen et al., 2023). While there is a study of introducing RAI in higher education teaching and practices (Aler Tubella et al., 2024), it does not extend to K-12 educational contexts. Most recently, Fu and Weng (2024) identified five key characteristics of RAI in education through synthesis of 40 K-20 studies; however, although they include a few studies of AI ethics education, the study have yet to examine the effects of the pedagogical designs for addressing these ethical principles.

2.4. The present study

According to previous sections, existing studies lack systematic examination of key pedagogical designs, assessment methods, and learning

outcomes specifically addressing K-12 AI ethics education. This gap leaves an incomplete understanding of how AI ethics is taught and assessed in K-12 settings, as well as how ethical learning outcomes contribute to students' developing AI literacy across cognitive, affective, and behavioral domains. To address these limitations, this study conducts a systematic review of empirical research on AI ethics education in K-12, addressing core RAI principles (Papagiannidis et al., 2025) and ABCE AI literacy framework (Ng, Wu, et al., 2024). By synthesizing existing work, this study seeks to overview the global trends of K-12 AI ethics education, identify effective practices and assessments, and highlight areas needing further development to advance AI literacy in alignment with RAI Principles. Specifically, this study seeks to answer the following research questions:

- 1. What are the key trends in the current empirical research on teaching and learning AI ethics in K-12 education settings, across various cultural and curricular contexts?
- 2. What are pedagogical designs for teaching and learning AI ethics in K-12 settings, addressing key responsible AI principle pillars?
- 3. What evaluative methods are employed to assess students' ethical learning across cognitive, affective, and behavioral domains?
- 4. What are the ethical learning outcomes across K-12 students' developmental stages, in terms of AI literacy development?

By addressing these questions, this review addresses the critical gap in examining K-12 AI ethics education. It advances AI literacy research by demonstrating how AI ethics progressively permeate all aspects of AI literacy development. The practical implications provide K-12 educators and policymakers with evidence-based insights for designing developmentally appropriate ethical learning experiences that foster responsible AI literacy.

3. Method

This review was carried out in four stages: data collection, data selection, data extraction, and data analysis, in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021).

3.1. Data collection

We searched four databases (i.e., Scopus, Web of Science, IEEE, ERIC) in March 2025 using Boolean-connected terms covering (1) AI and subdomains, (2) Ethics-related terms, (3) teaching and learning activities, and (4) K-12 education (see Table 1 for full terms).

3.2. Data selection

The study selection occurred in two stages: (1) Preliminary screening stage: we retained studies published between January 2014 and March 2025 (including pre-access) that were English-written, peer-reviewed

Table 1
The search terms.

Categories	Search terms
AI	("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Neural networks" OR "Natural Language Processing" OR "Generative AI" OR "Large language models")
Ethics	("Ethics" OR "Ethical issues" OR "Societal implications" OR "Social impact" OR "Socio-technical" OR "Socio-cultural")
Teaching & Learning activities	("Teaching" OR "Learning" OR "Education" OR "Curriculum" OR "Curricula" OR "Workshop" OR "Activities")
K-12 education	("K-12" OR "K12" OR "High school" OR "Secondary school" OR "Middle school" OR "Primary school" OR "Elementary school" OR "Kindergarten" OR "Preschool")

articles/conference proceedings related to AI or education, with duplicates removed; (2) Eligibility assessment stage: we applied the criteria outlined in Table 2 to examine the titles/abstracts, followed by full-text review, excluding review articles, opinion papers, and studies not addressing AI ethics teaching/learning in K-12 contexts. The final corpus comprised 68 studies, including 48 journal articles and 20 conference papers. The details of data collection and data selection are illustrated in the PRISMA diagram (see Fig. 2).

3.3. Data extraction

The extraction process occurred in two stages: (1) Demographic information (per-study analysis) capturing publication details (title, year, country/region, article type), educational context (grade levels, settings), and research methods; followed by (2) Detailed data extraction (per-case analysis) for some studies involving multiple teaching & learning cases, organized into five categories as specified in Table 3.

3.4. Data analysis

The analysis employed distinct coding schemes for each research question: (1) For RQ1, descriptive statistics and keyword frequency analysis identified key trends in K-12 AI ethics education research; (2) For RQ2, we first categorized the teaching and learning activities according to the seven pillars of RAI principles, then employed an inductive approach to identify the pedagogical designs that address each principle; (3) For RQ3, we categorized the assessment methods of AI ethics education used by researchers to evaluate students' performance in the cognitive, affective, and behavioral domains, based on Ng, Wu, et al. (2024)'s AI literacy framework; (4) For RQ4, deductive qualitative content analysis applied the same framework to trace ethical learning progression across all three domains.

The first and second authors independently reviewed and categorized the papers according to the established coding scheme. All discrepancies were resolved through discussion between the two coders, with inter-coder reliability of Cohen's Kappa coefficient (0.88), indicating the high-level of agreement between them. After completing the coding process, the findings were descriptively analyzed focusing on frequency, identified categories and exemplar excerpts.

4. Results

4.1. The key trends of current research in K-12 AI ethics education (RQ1)

This section analyzes publications on AI ethics education in K-12 settings, exploring yearly trends, global distribution, research methods, and mapping out the learning contexts with K-12 developmental stages. We highlight key patterns in the reviewed studies and compare the differences across various educational contexts.

Table 2
Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Empirical studies that involve K-12 students as participants.	Review studies Conceptual/Opinion paper Editorials/comments
Primarily focusing on teaching and learning AI ethics	Ethics of applying AI in Education Using GenAI to facilitate learning other subjects Discussing human-AI relationship in education
Formal/Informal K-12 educational settings	Higher Educational levels Adult Education Tertiary Education
Teaching and learning programs including AI ethics or social implications of AI	Survey studies without implementation AI education studies without any details related to ethical dimension

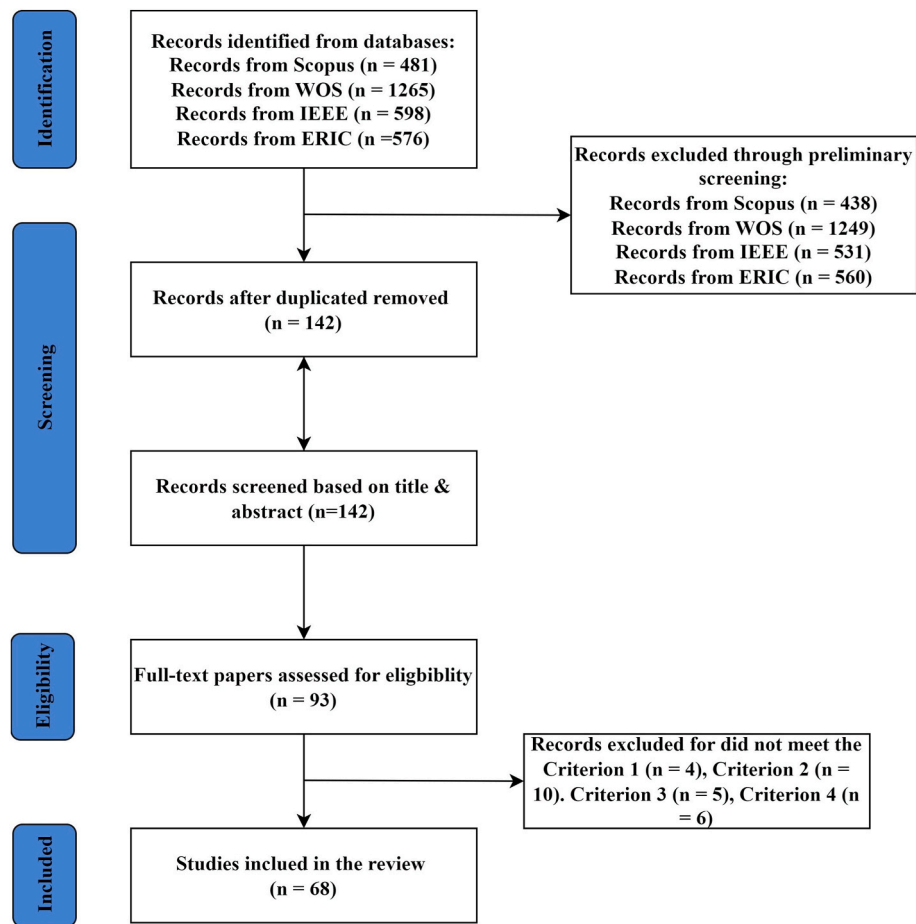


Fig. 2. PRISMA diagram.

Table 3
Detailed data extraction categories.

Research Design	Participants & Learning contexts	Pedagogical Designs	Assessment methods	Students' learning process
Specific research approaches employed	Sample size, time durations, & specific learning contexts	Specific teaching & learning activities	Assessment methods used to evaluate	Reported students' learning progression of the ethical learning activities

As illustrated in Fig. 3, research on AI ethics education in K-12 settings has grown steadily since its inception, reflecting increasing scholarly engagement with this emerging field. The first study appeared in 2019, marking the beginning of dedicated inquiry in this domain. Early contributions were sparse and predominantly conference papers. In 2020, the field expanded slightly with 1 journal article and 6 conference papers, followed by a modest increase in 2021 (3 journal articles and 7 conference papers). A pivotal shift occurred in 2022 when journal articles (n = 11) first outnumbered conference papers (n = 2), signaling the burst of interest. This trend continued in 2023 (15 journal articles and 1 conference paper) and 2024 (11 journal articles and 5 conference papers). Preliminary data of 2025 (up to March) includes 2 published journal articles and 3 pre-proofs, suggesting a sustained momentum in this critical area of research.

As illustrated in Fig. 4, the United States dominates the field with 28 of the 68 reviewed studies, reflecting its prominent role in shaping discourse on AI ethics education. In Asia, Hong Kong emerges as a regional leader with 12 studies, followed by China (5), South Korea (2), Japan (1), India (1), and Taiwan (1). European contributions are more evenly distributed, with Denmark (3), Finland (2), Spain (3), Greece (2), and smaller outputs from Ireland (2), Sweden (1), and the United

Kingdom (1). Nigeria stands out in Africa with 3 published studies, while the Middle East (Israel, 1) and Latin America (Brazil, 1) show early but limited engagement.

Regarding the research methods (see Table 4), 25 employed qualitative approaches to explore students' ethical learning process, offering in-depth insights into shifts in their perceptions and understandings of these topics. Quantitative methods (n = 8) were used to assess key constructs across three domains: (1) cognitive (e.g., AI concepts, critical thinking), (2) affective (e.g., AI ethical awareness, attitudes toward AI), and (3) behavioral (e.g., behavioral intentions). Most studies (n = 35 out of 68) adopted a mixed-methods approach (see Table 5), collecting both qualitative and quantitative data from multiple sources of students' learning outcomes of AI ethics education.

As detailed in Table 5, studies on AI ethics education are most extensively explored at the middle school level (n = 27 out of 68), where a significant imbalance exists between informal learning contexts (n = 19) and formal contexts (n = 8). In contrast, elementary, high school, as well as elementary school & middle school and middle school & high school levels exhibit a more balanced distribution of studies across formal and informal learning contexts, albeit with emerging research engagement. At the kindergarten level, research remains limited, with

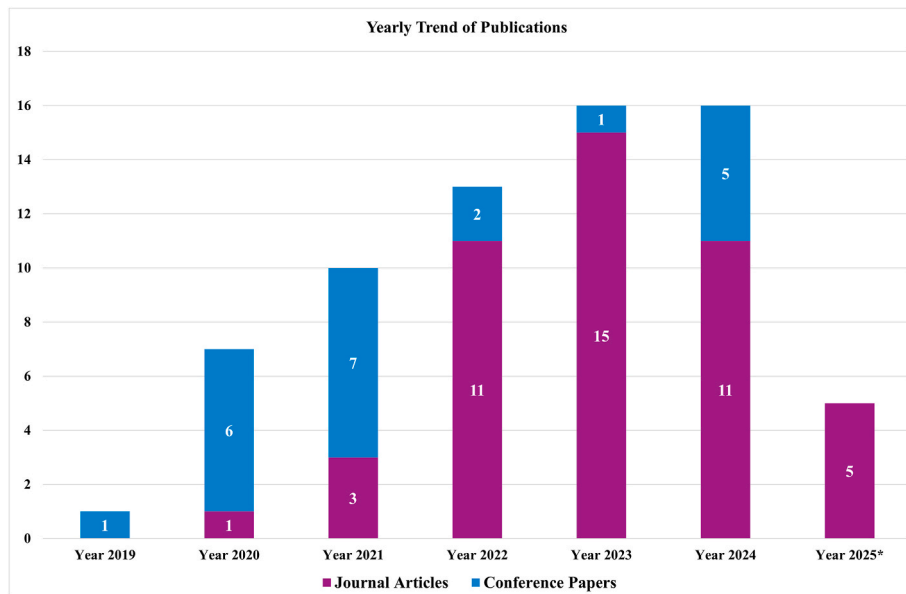


Fig. 3. Yearly trend of publications.

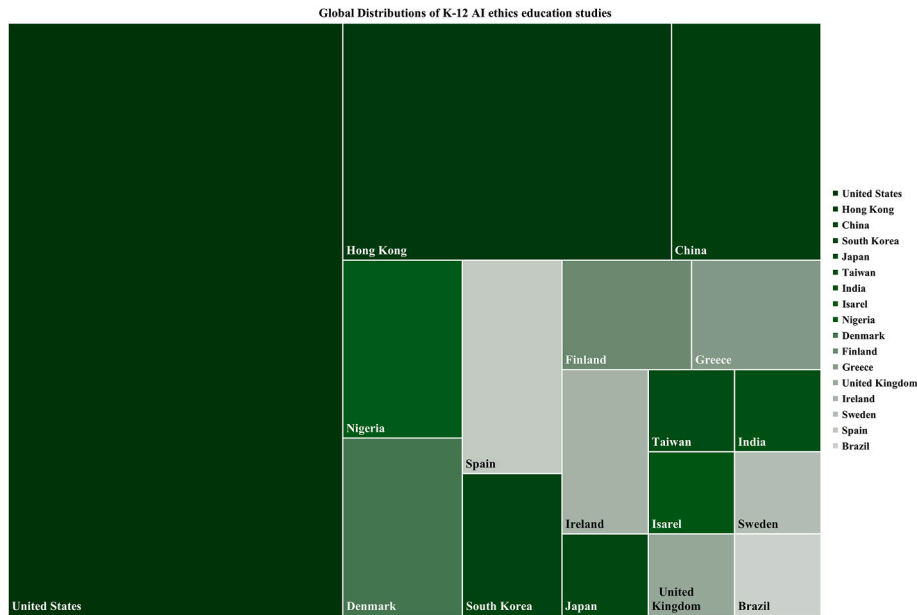


Fig. 4. Global distribution of AI ethics education studies.

only three studies identified. Notably, there is one study spanning across K-12 and university level, an emerging attempt to explore the usage of LLMs' impact on student understanding of AI ethics, across grade levels (Kajiwara & Kawabata, 2024).

4.2. The pedagogical designs addressed responsible AI principles (RQ2)

We analyzed 188 instances of teaching activities extracted from 68 studies, mapping pedagogical designs to the seven responsible AI principles across K-12 education. Through qualitative content analysis, we first deductively categorized these instances by the seven RAI principle pillars, then inductively identified 16 pedagogical designs from the instances (Fig. 5), revealing how these principles translate into developmentally appropriate practices. Subsequent sections detail each principle's associated pedagogical designs.

Accountability. Three major pedagogical designs were spotlighted

addressing *Accountability* (see Fig. 5). First, students were prompt to map responsibility in using AI (n = 8), particularly when they may cause harm. For instance, Sanusi, Omidiora, et al. (2023) had middle school students examine a case where their town's police department used facial recognition technology. Through ethical discussion, students explored the risks of unjust arrests and emphasized the deploying party's responsibility in addressing such concerns. However, Kajiwara and Kawabata (2024) identified a notable challenge for students in determining who should be responsible for using misinformation generated by LLMs. The second pedagogical design involved analyzing multiple stakeholders in the AI ecosystem (n = 5). Researchers used impact matrices to help students assess the roles and perspectives of different stakeholders in redesigning YouTube's recommendation algorithms (Pang et al., 2024; Williams et al., 2023; Zhang et al., 2023). Lastly, prompting accountability in AI prototyping (n = 4) engaged students in co-design activities. For example, Bilstrup et al. (2020) engaged high

Table 4
Research methods of included studies.

Research Methods	n	Studies
Mixed	35	Ali et al. (2019); Ali et al. (2021); Alonso (2020); Amplo and Butler (2023); Bendecheche et al. (2021); Chiu et al. (2021); Choi et al. (2024); Dai et al. (2024); Dipaola et al. (2020); Kahila et al. (2024); Kajiwaru and Kawabata (2024); Kim and Kwon (2024); Kong et al. (2023); Krakowski et al. (2022); Lee et al. (2021); Martins et al. (2024); Ng et al. (2022); Ng, Su, and Chu (2024); Norouzi et al. (2020); Shamir and Levin (2022); Su and Yang (2024a); Su and Yang (2024b); Van Brummelen et al. (2021); Zammit et al. (2022); Williams et al. (2023); Williams et al. (2024); Yim (2024b); Zhan et al. (2022); Zhang et al. (2023); Yue et al. (2025); Zhou et al. (2025); Kong et al. (2024); Ahuja and Kumar (2024); Zhao et al. (2024); Zhang et al. (2024)
Qualitative	25	Bilstrup et al. (2020); Burriss et al. (2024); Durall Gazulla et al. (2025); Henry et al. (2021); Irgens et al. (2022); Kaspersen et al. (2022); Kim et al. (2023); Kim et al. (2024); Lee et al. (2022); Relmasira et al. (2023); Reddy et al. (2021); Sabuncuoglu (2020); Sanusi, Omidiora, et al. (2023); Sanusi et al. (2024); Sanusi, Oyelere, et al. (2023); Schaper et al. (2020); Schaper et al. (2022); Skinner et al. (2020); Solyst et al. (2023); Ali, Ravi, Williams, Dipaola, and Breazeal (2024); Yang et al. (2024); Chang et al. (2025); Kim et al. (2025); Zhou et al. (2024); Zhou et al. (2025)
Quantitative	8	Fernández-Martínez et al. (2021); Lin et al. (2023); Oskotsky et al. (2022); Park and Kwon (2024); Xia et al. (2022); Melsión et al. (2021); Zammit et al. (2022); Dai (2024)

Table 5
Sums of learning contexts and grade levels.

Grade Levels	Formal	Formal & Informal	Informal
Across K-12	0	0	1
Elementary School	8	1	7
Elementary & Middle	2	0	1
Kindergarten	1	0	2
Middle School	8	0	19
Middle & High	5	0	2
High School	5	0	6

school students to responsibly reflect on how their choices of AI algorithms and datasets could lead to unintended ethical or social consequences.

Transparency. Two major pedagogical designs were identified addressing *Transparency* (see Fig. 5). First, the AI algorithm demystification ($n = 22$) involves using various pedagogical methods to reveal the ‘black box’ nature of AI decision-making. For example, the Neural Network game had students role-play as nodes in a network, passing information through layers to understand node, link, and layer interactions (Lee et al., 2021; Zhang et al., 2023). Dai et al. (2024) and Zhou et al. (2025) both utilized analogy-based approaches, comparing human-AI decision-making and using bee pollination metaphors respectively, to explain how AI algorithms select similar data patterns. The second design ($n = 14$) is about explainable AI (XAI) techniques (e.g., Alonso, 2020; Martins et al., 2024; Ng et al., 2022). For instance, Melsión et al. (2021) used Grad-CAM, a visualized explainability technique, to demonstrate how machine learning algorithms can amplify gender bias, enabling students to critically evaluate AI systems.

Diversity, Non-discrimination, & Fairness. Four major pedagogical designs addressing this principle in AI education were identified (see Fig. 5). The most common design was bias origins analysis ($n = 25$), where students interacted with tangible tools like Scratch and Google Teachable Machine to explore how dataset biases affect outcomes - for example, Ali et al. (2019) presented an Google Teachable Machine

activities, showing middle school students how biased animal datasets influence AI results. The second design focused on bias identification in AI applications ($n = 16$), with pedagogical practices that guided students to analyze daily-use platforms like YouTube, or Spotify. For example, high school students recognized image generators (i.e., DALLÉ-2) disproportionately represented ‘pretty girls’ as young, white blondes (Ali et al., 2024). The third design involved bias mitigation ($n = 11$), featuring structured exercises where students learned to address gender and racial biases through dataset curation (e.g., Alonso, 2020; Zhou et al., 2025). Finally, the fairness audits ($n = 3$) encourage students to design fair AI systems. For example, Skinner et al. (2020) demonstrated a project-based learning where students assessed and redesigned an AI librarian system to examine fairness trade-offs.

Societal and Environmental Well-being. Two major pedagogical designs were identified addressing this principle (See Fig. 5). Most studies ($n = 25$) guided students in evaluating AI’s societal impacts, examining systems like self-driving cars, such as decision-making for self-driving cars (Choi et al., 2024), AI generated media by Deepfake (Durall Gazulla et al., 2025). For example, activities like the ‘Unanticipated Consequences’ exercise involved students to reflect on how poorly designed AI systems can lead to significant negative impacts and potential of lawsuits (Lee et al., 2021). Similarly, Lin et al. (2023) used AR applications to facilitate dilemma discussions about automated road trollers’ effects on employment and local economies. Another ($n = 6$) focusing on designing AI for social or environmental good. Burriss et al. (2024) had middle school and high students role-play as policymakers analyzing gaps for teenagers in the White House Blueprint for an AI Bill of Rights. Both Kim and Kwon (2024) and Ng, Su, and Chu (2024) engaged elementary and middle students to build recycle bins based on AI selection.

Technical robustness & safety. A few studies addressed this principle through two pedagogical designs (see Fig. 5). The majority ($n = 4$) focused on spotting AI-generated misinformation. For example, middle school students analyzed how Deepfake celebrity images on social media can mislead audiences, emphasizing the need to evaluate the credibility of AI-generated content (Ali et al., 2021). The second design involved conducting reliability checks for AI systems. For example, students assessed COVID-19 information chatbots created with Microsoft Azure QnA Maker, highlighting the importance of expert validation (Kong et al., 2024).

Privacy & Data governance. Two pedagogical designs addressing Privacy & Data governance were highlighted (see Fig. 5). The first focused on raising awareness of privacy risks. For example, students critically examined why their schools implemented AI-powered screen monitoring software that tracked activities beyond school hours without consent (Burriss et al., 2024). Lee et al. (2022) and Bendecheche et al. (2021) asked students to evaluate the privacy settings and data trade-offs in popular AI-based applications like YouTube and Spotify. The second design engaged students in mitigating risks of data breaches ($n = 4$). For example, Chang et al. (2025) involved high school students in co-designing a classroom collaboration AI application, establishing an ethical privacy framework through iterative prototyping discussions. Kim and Kwon (2024) guided elementary students training AI models using non-sensitive gesture data, avoiding personal data privacy concerns.

Human Agency & Oversight. A distinctive pedagogical design addressing this principle is human-in-the-loop design ($n = 6$) (see Fig. 5). For example, Schaper et al. (2020) involved elementary students explore robot limitations in interpreting cultural diversity through scenarios involving robot presidents, highlighting the need for human-programmed rules. Yue et al. (2025) encouraged elementary students to recognize if AI-generated contents need human verification. Similarly, Zhou et al. (2025) engaged middle school students in co-designing AI systems human feedback buttons for flagging inappropriate AI-generated content or recommendations.

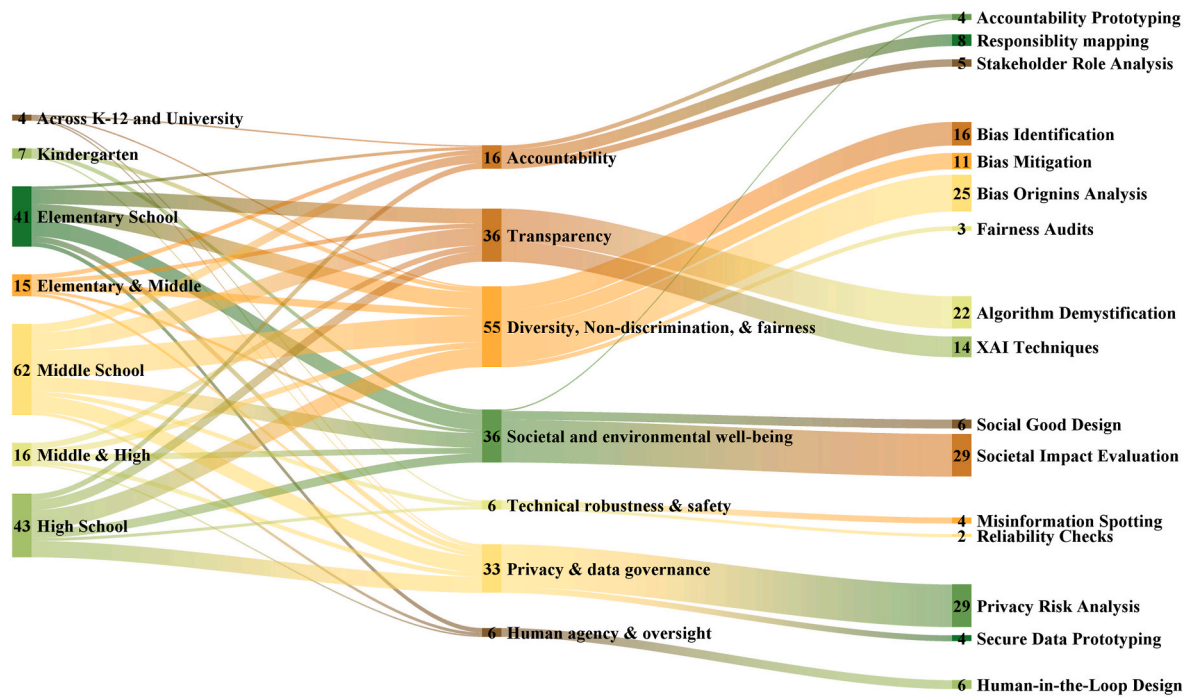


Fig. 5. Map of K-12 pedagogical designs addressing responsible AI principles.

4.3. Assessment methods to measure AI ethics development (RQ3)

We categorized methods for evaluating ethical AI learning into three domains: cognitive, attitudinal, and behavioral (Ng, Wu, et al., 2024), including both quantitative and qualitative methods. For cognitive domain, researchers focused on assessing students' cognitive understanding of AI ethics. For example, Lee et al. (2021) and Zhang et al. (2023) who administered the AI Concept Inventory (AI-CI) to evaluate students' comprehension of AI concepts and their ethical and societal implications, and triangulated the results of open-end survey and semi-structured interviews. Martins et al. (2024) applied a scoring rubric to evaluate student-developed machine learning models based on criteria of originality and design appropriateness, with analyzing the semi-structured interviews of students' reflections on the design process and their evolving understanding of machine learning concepts. Lin et al. (2023) employed higher-order thinking tendency questionnaire, which include five key dimensions: *creativity tendency*, *team cooperation tendency*, *communicative tendency*, *metacognitive awareness*, *complex problem-solving tendency*. Ng et al. (2022) analyzed the narratives of the digital stories created by students to showcase their understanding of AI and its societal implications.

For affective attributes, researchers examined shifts in students' attitudes, values, or perceptions toward AI. The most important construct is ethical awareness, which refers to students' values and attitudes toward AI ethical issues, including their ability to identify, assess, and respond to these issues responsibly (Stahl, 2021). For example, Kong et al. (2023) used a pre- and post-ethical awareness survey with 12 items, guided by ethical principles of *human autonomy*, *beneficence of AI*, and *fairness of AI*, whereas Dai et al. (2024) assessed ethical awareness by employing scenario-based multiple-choice questions, requiring students to analyze real-life situations and make ethical decisions. In another study, Dai (2024) measured the change of students learning attitudes, using three questionnaires: AI learning confidence (AICF), intrinsic motivation to learn AI (AIIM), and AI learning anxiety (AIAX). Lee et al. (2021) administered AI career futures survey to explore students' awareness of AI-related careers, the skills required for those roles, and their career motivations in STEM fields. Kajiwar and Kawabata

Table 6

Assessment methods employed by researchers.

Assessed Changes	Research Methods	Examples	Sample studies
Cognitive domains	Quantitative	- AI concept inventory (AI-CI) - Scoring Rubric - Higher-order thinking tendency questionnaire	Lee et al. (2021); Zhang et al. (2023); Martins et al. (2024); Lin et al. (2023).
	Qualitative	- Student-created artifacts - Open-ended survey - Semi-structured interviews	Ng et al. (2022); Irgens et al. (2022); Martins et al. (2024); Zhang et al. (2023); Kim et al. (2023)
Affective domains	Quantitative	- Ethical awareness - Learning Attitudes - Self-efficacy/confidence - Technology acceptance - Career interest in AI - AI for social good	Kong et al. (2023); Dai et al. (2024); Dai (2024); (Yue et al., 2025) Kim and Kwon (2024); Kajiwar and Kawabata (2024); Lee et al. (2021); Kim and Kwon (2024)
	Qualitative	- Reflection paper - Art creations - Group discussions	Bilstrup et al. (2020); Yim (2024); Ali et al. (2024); Durall Gazulla et al. (2025).
Behavioral domains	Quantitative	- Lag Sequential analysis - Behavioral Intentions - Dispositional autonomy	Lin et al. (2023); Kim and Kwon (2024); Zhou et al. (2025)
	Qualitative	- Observation of design - Embodied Activities	Kaspersen et al. (2022); Pang et al. (2024); Kim et al. (2025); Zhou et al. (2025)

(2024) adopted a questionnaire-based survey to measure students' acceptance of LLMs, focusing on their perceptions of *social impact*, *performance expectancy*, *effort expectancy*, and *attitudes towards LLM use*. Kim and Kwon (2024) measured students' self-efficacy and motivation for use AI for social good. In terms of qualitative methods, researchers analyze the aesthetics, symbolism, and choices made during the design process can uncover underlying values and attitudes of students towards AI ethics (e.g., Bilstrup et al., 2020; Durall Gazulla et al., 2025; Yim, 2024a). For example, Ali et al. (2024) analyzed students' discussions of the AI-generated images they created depicting their dream jobs, along with their reflections on how these tools might transform future creative professions.

Regarding behavioral domains, researchers focus on analyzing changes in students' behaviors or behavioral intentions. For example, Lin et al. (2023) employed lag sequential analysis (LSA) to examine student behavioral patterns in the ethical dilemma-based discussion activity, which helps track and identify key sequences of actions or responses that contributed to effective ethical learning. Kim and Kwon (2024) assessed students' AI readiness and behavioral intentions in learning AI before and after curriculum implementation of AI curriculum with tangible tools. Notably, Zhu et al. (2025) measured changes in children's dispositional autonomy, which represents their intentional inclinations to determine their own actions when interacting with AI systems. Kaspersen et al. (2022) recorded students' discussions during collaborative design activities, analyzing critical incidents related to the application of machine learning concepts during the design process. Kim et al. (2025) and Zhou et al. (2025) recorded and analyzed students' interactions of embodied learning activities in stimulating AI algorithms.

4.4. Students' ethical learning outcomes (RQ4)

Our analysis of learning outcomes from 68 studies across diverse contexts reveals that ethical learning outcomes manifest through three dimensions: cognitive gains (e.g., understanding algorithmic fairness), affective shifts (e.g., increased ethical awareness), and behavioral changes (e.g., implementing accountable designs). These manifestations can be further mapped across progressive competence levels of AI literacy development: from knowing & understanding AI to applying & using AI, and ultimately to evaluating & creating. Representative examples are detailed in Table 7.

Knowing & Understand AI. After receiving AI ethics education, students developed a fundamental understanding of AI's socio-technical nature. In the cognitive domain, students shifted from perceiving AI as inherently objective to recognizing its potential biases, data misuse, and the importance of ethical considerations (Kim et al., 2023). In the

affective domain, Dai et al. (2024) found that an analogy-based approach of teaching fundamental concepts of AI significantly improved upper elementary students' AI ethical awareness ($p < 0.05$, Cohen's $d = 0.95$). In a subsequent study, she reported that dual-contrast approach significantly improved girls' ethical awareness in the experimental group, $F(1, 143) = 30.26$; $p < .001$, $\eta_p^2 = .18$, though no significance difference with boys $F(1, 143) = 1.49$; $p = .22$, $\eta_p^2 = .01$ (Dai, 2024). Kong et al. (2023) claimed that secondary students have difficulty in understanding abstract ethical principles comparing to university students in their previous studies. After they modified their teaching methods with project-based learning, they reported a significant increase in secondary students' comprehension of AI ethics ($M_{diff} = 0.09$, $Z = -2.73$, $p < 0.01$) (Kong et al., 2024). In the behavioral domain, Pang et al. (2024) observed that female students of color were particularly active in identifying and discussing bias in AI systems. Solyst et al. (2023) found that girls in their workshop readily pinpointed potential harms from digital assistant technology, whereas most children aged 7–11 who typically prioritize playing with AI over considering technical drawbacks. Kim et al. (2025) demonstrated an episode of students' embodied learning activities, where the disagreements between "AI agent" and "labeler" roles exposed inherent biases in classification tasks when distinguishing student hair colors.

Using & Applying AI. Students demonstrated the competency of responsible and ethical use of AI tools, considering ethical data usage. In the cognitive domain, Kajiwarra and Kawabata (2024) reported that around 70 % of students understood the importance of proper data management and minimizing harm when using ChatGPT. Yue et al. (2025) demonstrated that students reflected on the consequences and social impact of using AI, especially in the scenarios involving image recognition, with questioning the accuracy. In the affective domain, they also revealed that students held more rational and neutral attitudes toward AI, in terms of certain items, such as "AI system can outperform human" (from 3.50 to 3.02), "I shiver with discomfort when I think about future uses of AI" (from 3.10 to 2.67). Kim and Kwon (2024) reported that students' self-efficacy in learning AI ($F[1, 112] = 28.85$, $p < 0.001$, $\eta_p^2 = 0.21$) and motivations to use AI for social good ($F[1, 112] = 20.45$, $p < 0.001$, $\eta_p^2 = 0.15$) significantly improved after using tangible computing tools. In the behavioral domain, Ali et al. (2021) found no significant change in the survey of middle school students' ability to detect AI-generated pictures after the Spot the Deepfakes activity (54.68 to 53.87, $p = .488$), whereas students highlighted impersonation and misinformation as key harms, with one student suggesting government regulation to counter Deepfakes.

Evaluate & Create AI. Students developed the competency to

Table 7
Examples of AI ethics learning outcomes in K-12 education.

AI literacy competencies	Cognitive outcomes	Affective outcomes	Behavioral outcomes
Knowing & Understanding AI	Students demonstrated an increased understanding of AI's socio-technical nature, recognizing that AI is not inherently objective but reflects the biases of its creators and the training datasets (Kim et al., 2025; Sanusi, Oyelere et al., 2023; Yim, 2024b).	Students' ethical awareness significantly increased after learning fundamental AI concepts, as exemplified by human-AI comparisons and analogies discussions (Dai, 2024; Dai et al., 2024).	Students actively participated in or group discussions, or embodied activities particularly the ones who are from underrepresented groups (Kim et al., 2025; Pang et al., 2024; Solyst et al., 2023).
Using & Applying	Students understood the importance of proper data management and the potential harms, such as, bias of generated contents, misinformation, issues of plagiarisms when using generative AI (Ali et al., 2024; Kajiwarra & Kawabata, 2024; Yue et al., 2025).	Students' self-efficacy in learning AI and motivations for using AI for social good significantly improved, as exemplified by their engagement in manipulating tangible computing tools (Kim & Kwon, 2024).	Students made their commitments to responsible AI use through their choices and actions by actively recognizing AI-generated misinformation on social media (Ali et al., 2021).
Evaluating & Creating AI	Students showed an increased AI ethical reasoning capability and higher-order thinking tendency after participating AR-based dilemma-based discussion of real-world examples. (Lin et al., 2023).	Students demonstrated increased career interest in AI-related careers, as becoming responsible creators of accountable AI systems for human well-being (Pang et al., 2024; Park & Kwon, 2024).	Students conducted audits of AI systems they designed, evaluate datasets for representativeness, and propose solutions and reducing potential ethical issues (Alonso, 2020; Bilstrup et al., 2020; Yue et al., 2025).

critically evaluate AI systems and design AI with ethical principles. In cognitive domain, Lin et al. (2023) found that elementary students in AR-based ethical dilemma discussions showed greater higher-order thinking skills tendency, scoring significantly higher on post-tests than the mobile app control group ($F = 4.420, p = .004, \eta_p^2 = 0.05$). In the affective domain, students are motivated to pursue AI-related careers to address AI ethics issues. For example, Park and Kwon (2024) reported that their AI education program for South Korean middle school students, focused on AI's social impact, ethics, physical computing, and problem-solving, led to significant increases in students' "interest in technology" (3.96 to 4.17, $p < 0.05$) and "career aspirations in technology" (3.83 to 4.12, $p < 0.01$). In the behavioural domain, students actively engaged in evaluating AI systems and their design decisions with a critical perspective. For example, Kaspersen et al. (2022) observed that students questioned the fairness and accuracy of predictions based on limited data and the potential for bias of the results produced by VoteStratesML for democratic elections. Kahila et al. (2024) also observed that students critically test classifiers, explain observations, and iteratively refine apps—enabling them to analyze model behavior, identify training data biases, and understand data-to-output relationships for responsible AI development. Yue et al. (2025) reported that four students critically recognized potential malicious use risks in their AI product designs, by bad people like poachers.

5. Discussion

By further synthesizing the findings, this discussion examines four prominent gaps identified in the current empirical research of K-12 AI ethics education: (1) East-West contextual disparities, (2) Gaps in addressing RAI principles and emerging LLMs' ethical challenges, (3) methodological limitations, (4) Issues in assessing ethical learning outcomes.

5.1. East-West contextual disparities of K-12 AI ethics education

Our systematic comparison reveals distinct cultural approaches to AI ethics education between Western and Eastern contexts. In Western settings, particularly the U.S. and Europe, AI ethics education predominantly occurs in informal learning contexts such as after-school workshops and summer school programs, often emphasizing critical engagement with specific ethical issues like bias, privacy, and societal well-being (e.g., Bendecheche et al., 2021; Chang et al., 2025; Durall Gazulla et al., 2025; Lee et al., 2022). This approach is exemplified by students identifying how generative AI applications perpetuate beauty standards by defaulting to white, blonde features for "pretty girls" (Ali et al., 2024), as well as projects where African American students develop community-specific applications to address local inequities (Pang et al., 2024). Such activities often involve ethnically diverse participants to mirror multicultural demographics (e.g., Lee et al., 2021; Zhang et al., 2023).

By contrast, Eastern education systems, particularly in Hong Kong, China, and South Korea, typically integrate AI ethics into formal classroom curricula as part of holistic AI literacy frameworks (e.g., Chiu, 2021; Dai, 2024; Kong et al., 2023; Yue et al., 2025). Here, the focus shifts toward broader societal impacts, examining issues like algorithmic fairness in food delivery platforms (Zhu et al., 2025) or ethical dilemmas for autonomous vehicles (Lin et al., 2023)—topics that resonate with these regions' rapid AI adoptions. Student projects often prioritize social benefits in a broader sense, as seen in environmental well-being initiatives like AI recycling bins (Kim & Kwon, 2024; Ng, Su, & Chu, 2024). The disparities between eastern and western contexts suggest a culturally responsive design to approach K-12 AI ethics education in the future research and curriculum development (Eguchi et al., 2021).

5.2. Gaps in addressing RAI principles and emerging LLMs' ethical challenges

Our analysis also reveals that some critical responsible AI principles lack addressing in K-12 education levels. While *Diversity, Non-discrimination, Fairness and Societal/Environmental Well-being* dominate the K-12 AI ethics curricula, *Technical Robustness & Safety* receive scant attention. As Kajiwaru and Kawabata (2024) noted that younger students are more likely to rely on information generated by ChatGPT making them susceptible to misinformation without proper safeguards. To mitigate these risks, it is imperative to incorporate reliability checks and critical evaluation skills through AI ethics education from an early age.

Moreover, *Human Oversight & Agency* in AI systems require greater attention. While a few studies have explored human-in-the-loop features in design activities (e.g., Yue et al., 2025; Zhu et al., 2025), a fundamental understanding of key mechanisms, such as how reinforcement learning with human feedback works in LLMs (González Barman et al., 2025), remains essential. This knowledge could help children demystify the inner workings of these complex systems, fostering greater transparency, trust, and agency in their interactions with LLMs.

Furthermore, *Accountability* need to be further strengthened. Current studies primarily address *Accountability* for designers and deployers of AI applications (e.g., Bilstrup et al., 2020; Zhang et al., 2024), whether through stakeholder analysis or responsibility mapping, but neglect the responsibilities of end-users. Only a few studies have briefly noted the accountability of students using LLMs (e.g., Ali et al., 2024; Kajiwaru & Kawabata, 2024), yet without further discussing how they should uphold ethical principles in their responsible usage. This gap is critical to address, as students' accountability directly impacts their academic integrity and the ethical adoption of AI in education (Han et al., 2025; Meça & Shkëlzeni, 2024).

Additionally, a few studies noted the emerging issues of data ownership and copyright concerns brought by ubiquitous AI-generated content (AIGC) on social media. For instance, students agreed that all stakeholders—from data providers and AI developers to end-users—should have ownership rights over AIGC and exercise detailed audit control, but they disagreed on how to allocate ownership, with some advocating for user control and others emphasizing stricter human oversight of content (Zhu et al., 2025). Ali et al. (2024) observed that while using GenAI to envision their career futures, students also expressed concerns that AI-generated content could threaten creative jobs. These emerging issues aroused by LLMs-based GenAI need to be further addressed by the future studies.

5.3. The limitations of methodological approaches

Most studies of AI ethics education collect data from multiple data sources, evaluating changes before and after implementing the curriculum, yet remains primarily focused on immediate cognitive and affective outcomes (e.g., conceptual knowledge gains, attitude/values shifts) (e.g., Dai, 2024; Kong et al., 2023; Lin et al., 2023). While emerging studies analyze behavioral data during learning activities, such as embodied role-playing games or project-based design tasks (e.g., Chang et al., 2025; Kim et al., 2025; Zhou et al., 2025), a critical gap still persists: none of longitudinal studies exist to examine how these AI ethics interventions impact students' long-term ethical awareness, critical thinking, and sense of responsibility when interacting with AI systems outside of educational contexts. This gap is especially critical because children encounter ubiquitous AI everywhere, such as navigating social media recommendation systems, using generative AI for homework, or interacting with customer service chatbots, which present more complex and ambiguous ethical challenges than controlled classroom activities.

5.4. The issues of assessing ethical learning outcomes

The current state of AI ethics education research suffers from significant heterogeneity in assessing learning outcomes, and the varied contexts of implementation. This diversity makes meaningful meta-analyses nearly impossible. For example, studies by Dai et al. (2024) and Zhou et al. (2025), both employing analogical learning approaches, assessed different domains—cognitive and affective gains versus behavioral learning engagements—rendering their findings incomparable in terms of pedagogical efficacy. Addressing these issues requires a comprehensive strategy that incorporates both qualitative and quantitative methods to capture a holistic understanding of how individuals perceive and act on AI ethics.

Researching and measuring AI ethics effectively faces challenges in integrating diverse methodologies. While critics often argue that quantitative methods are at odds with ethical deliberation, tools like questionnaires and surveys can facilitate self-assessment and reflection, yet they require careful design to ensure meaningful insights into ethical decision-making. Instruments such as the AI Concept Inventory (Lee et al., 2021) and the Ethical Measurement of AI Literacy (Ng, Wu, et al., 2024) offer ways to quantitatively assess students’ self-efficacy in AI ethics and track changes over time. However, it is crucial to recognize the limitations of these tools and the need to complement them with evaluations of how individuals exercise moral reasoning and decision-making processes.

Ethical reasoning involves predicting outcomes and examining motives, which are essential for understanding moral responsibilities. The challenge lies in effectively exploring students’ ethical development by analyzing projects, observations, and artifact-based interviews to reveal how they embed values like respect, transparency, and fairness into their learning processes. This requires a structured method to assess and enhance both students’ self-efficacy and ethical behaviors, ensuring that researchers and educators can understand how students perceive their ethical reasoning and decision-making, and how those perceptions translate into action.

6. Implications

To address the issues addressed in 5.4 section, we propose reconceptualizing AI ethics as a transformative competency within ABCE AI literacy frameworks, structured to: (1) establish ethics as a core pillar, rather than an ancillary topic in AI curricula; (2) assess ethical learning outcomes across cognitive, affective, and behavioral domains; and (3) sequence ethical learning to progressively address key responsible AI principles along a developmental continuum—from foundational understanding, such as recognizing bias origins, to advanced skills, like building equitable AI applications. This approach highlights the unique role of AI ethics in AI literacy development and facilitates rigorous, comparable evaluations of educational interventions across various contexts.

By integrating AI ethics as a transformative dimension across the learning domains, the framework ensures that ethical considerations are embedded throughout AI literacy development, enabling students to engage with responsible AI principles while acquiring technical knowledge and skills. As depicted in Fig. 6, the ethical dimension permeates the cognitive, affective, and behavioral domains, infusing them with human-centered values while preserving their distinct roles. Detailed learning outcomes and competency development aligned with ethical learning are mapped out in Table 8, illustrating how these principles manifest across the three domains.

For educators, this competency-based responsible AI literacy framework provides practical pathways to embed ethical learning into K-12 AI curricula. For instance, teachers can introduce ethical concepts such as algorithmic bias, data privacy, and transparency through analogy-based pedagogies, using embodied activities to help students grasp the data practices underlying AI mechanisms (e.g., Dai, 2024;



Fig. 6. The Responsible AI literacy Framework.

Table 8
Competency-oriented responsible AI literacy framework.

Key Dimension	Cognitive	Affective	Behavioral
Know & Understand AI from a socio-technical Perspective	Develop a fundamental understanding of AI's socio-technical nature, including ethical concepts like algorithmic biases and explainability.	Develop a fundamental understanding of AI's socio-technical nature, including ethical concepts like algorithmic biases and explainability.	Articulate informed perspectives on AI's societal role; actively engage in discussions or activities on AI ethical challenges.
Use & Apply AI in a responsible manner	Understand accountability, ethical considerations, risks, and benefits of AI deployment.	Foster responsibility and empathy for individuals/groups affected by AI applications.	Make ethical decisions when using AI; demonstrate accountability in AI interactions.
Evaluate & Create AI Systems with a critical lens	Develop higher-order thinking skills; analyze societal implications, resolving dilemmas).	Commit to ethical design principles; maintain a critical mindset toward AI impacts.	Create AI for social good, designing responsible AI systems that address biases and inequities.

Zhou et al., 2025). Role-playing games can be employed to encourage students to critically reflect on their personal responsibilities when interacting with AI systems in their daily lives, promoting empathy and ethical awareness (e.g., Ali et al., 2021; Kim et al., 2025). Design-based activities can be structured as collaborative learning experiences, where students work together to integrate ethical principles into the creation of their artifacts, thus enhancing critical thinking and teamwork skills (e.g, Chang et al., 2025; Williams et al., 2024). The framework supports flexible learning assessments across cognitive, affective, and behavioral domains, enabling educators to tailor evaluations to specific learning objectives and track students’ ethical development accordingly.

Education policymakers are crucial in expanding access to responsible AI ethics literacy. Key priorities should focus on: (1) Updating K-12 education standards to explicitly incorporate AI ethics benchmarks at

each grade level, ensuring core responsible AI principles are integrated into the AI curricula; (2) Designing adaptive policy frameworks with students and teachers to ensure these resources effectively address their specific needs and ethical challenges (e.g., Ali et al., 2021; Burriss et al., 2024); and (3) Establishing educational programs to assist schools in marginalized communities in accessing AI ethics tools and teacher training, ensuring equitable and inclusive opportunities for all students (e.g., Chiu, 2021; Williams et al., 2024; Kim et al. 2025).

Ultimately, the framework empowers K-12 students not only not only enhances students' cognitive understanding of AI ethics but also equips them with affective attributes and behavioral engagement needed to navigate the ethical challenges of the AI-driven society. It develops their capacity for ethical decision-making whether they engage with AI as informed citizens, responsible end-users, or accountable designers. Together, these interconnected competences cultivate responsible AI literacy—preparing K-12 students to build futures where technology advances both economic prosperity and human well-being.

7. Future research directions

First, future research can address the emerging ethical challenges aroused by LLMs-based GenAI technologies (e.g., Ali et al., 2024; Kajiwara & Kawabata, 2024; Zhu et al., 2025). While issues like academic dishonesty have gained increasing attention in higher education (Yusuf et al., 2024), its exploration within K-12 education, especially in relation to AI ethics education, remains limited. The inherent complexity of LLMs further complicates efforts to address responsible AI principles including *Accountability*, *Transparency* and *Human Oversight & Agency*. Future research can also explore developmentally appropriate pedagogical designs to help K-12 students understand LLM functionality and emerging ethical issues to teach these critical principles.

The second research direction is developing students' ability to create fair and accountable AI systems. Studies show students frequently struggle to consider diverse perspectives beyond their immediate experiences, limiting their capacity to design inclusive technologies (Williams et al., 2024). Some students also tend to shift ethical responsibility to end-users through oversimplified solutions (Bilstrup et al., 2020). To address this, researchers should adopt empathy-driven approaches that highlight real-world societal impacts, fostering both designer accountability in AI development and student awareness of end-user responsibilities.

Another emerging area of focus involves engaging K-12 students directly in AI policy analysis and development. As AI governance frameworks continue evolving, there's growing recognition that K-12 students can participate in shaping the policies that will affect their digital futures (Ali et al., 2021; Burriss et al., 2024). This approach cultivates both critical thinking skills and their dispositional autonomy in AI governance (Zhu et al., 2025). By creating structured opportunities for students to analyze, critique, and contribute to policy discussions, researchers can empower the next generation to advocate for AI systems that align with human values.

8. Conclusion

In conclusion, this systematic review elucidates AI ethics' pivotal role in shaping responsible AI literacy for K-12 students. We first mapped out global trends in current K-12 AI ethics education research. Then, we examined the teaching and learning activities addressing even major responsible AI principles pillars, highlighting 16 effective pedagogical designs of K-12 AI ethics education. We also analyzed the methodological approaches and instruments of the included studies. Finally, we mapped out students' ethical learning outcomes across cognitive, affective, and behavioral domains.

By further synthesizing the results, we discussed the disparities of eastern and western disparities, gaps in addressing responsible AI principles and emerging ethical issues brought by LLMs-based GenAI,

limitations of methodological approaches, and issues in assessing ethical learning. Building on the synthesis, we proposed a competency-based responsible AI literacy framework that regards ethical learning as a transformative dimension for nurturing responsible digital citizens with higher-order thinking, to inform K-12 educators and policymakers. We also suggest future research directions to address ethical issues of emerging AI technologies, enhance students' skills and empathy in creating fair AI systems, and involve K-12 students in AI policy analysis and development.

CRedit authorship contribution statement

Ming Ma: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Davy Tsz Kit Ng:** Writing – review & editing, Formal analysis, Conceptualization. **Zhi-chun Liu:** Writing – review & editing, Supervision, Methodology. **Gary K.W. Wong:** Supervision.

Statements on open data and ethics

This systematic review is based on data extracted from publicly accessible databases and scholarly publications. All sources are appropriately acknowledged in the reference section, with detailed citations for each included study. Ethical approval and consent were not required, as this research involves a comprehensive analysis of existing literature.

Declare of use generative AI

During the preparation of this work the authors used Deepseek-R1 model to improve the readability. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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